

Due October 6 at 3:00 pm

(6 questions, 125 points total)

1. (15 pts.) Boolean functions as propositional formulas

Write propositional formulas for the following Boolean functions:

- (a) The majority function $M(x_1, x_2, x_3)$, which is true if and only if a majority of its inputs are true.

Answer:

$$M(x_1, x_2, x_3) = (x_1 \wedge x_2) \vee (x_1 \wedge x_3) \vee (x_2 \wedge x_3) \quad (1)$$

- (b) The if-then-else function $ITE(i, t, e)$, which is equal to t if $i = \top$ and otherwise is equal to e (this also called a multiplexer; here we're looking at a simple version where the inputs t and e are a single bit).

Answer:

$$ITE(i, t, e) = (i \wedge t) \vee (\neg i \wedge e) \quad (2)$$

- (c) The threshold function $T_{\leq 5}(x_1, x_2, x_3, x_4)$, which is true if and only if the integer represented in binary as $x_1x_2x_3x_4$ is at most 5 (e.g. $T_{\leq 5}(1, 1, 0, 1) = \perp$ since $1101_2 = 13 \not\leq 5$).

Answer:

$$T_{\leq 5}(x_1, x_2, x_3, x_4) = \neg x_1 \wedge (\neg x_2 \wedge \neg x_3) \quad (3)$$

2. (20 pts.) Satisfiability and validity

For each of the following formulas, state whether it is satisfiable and/or valid.

- (a) $x \rightarrow \neg x$

Answer: simplifies to $x \vee \neg x$ which is a tautology and therefore satisfiable and valid.

- (b) $(x \vee y) \wedge (\neg x \vee y) \wedge (x \vee \neg y) \wedge (\neg x \vee \neg y)$

Answer:

- (c) $(x \rightarrow y) \leftrightarrow (\neg x \vee y)$

Answer:

- (d) $x \rightarrow (\neg x \rightarrow (y \wedge \neg y))$

Answer:

- (e) $(x \vee \neg x) \wedge (x \vee y \vee z)$

Answer:

3. (20 pts.) Equivalence and equisatisfiability

For each of the following pairs of formulas, state whether they are equivalent and/or equisatisfiable.

(a) x and y

Answer:

(b) $x \leftrightarrow y$ and $(x \rightarrow y) \wedge (\neg x \rightarrow \neg y)$

Answer:

(c) $(x \vee y \vee z) \wedge (x \vee \neg z)$ and $x \vee y$

Answer:

(d) $x \rightarrow \neg x$ and $\perp$

Answer:

(e) $(x \vee y) \wedge \neg y \wedge \neg x$ and $\neg(x \vee \neg x)$

Answer:

4. (20 pts.) Conversion to conjunctive normal form (CNF, a.k.a. product-of-sums)

Consider the formula $\neg((x \vee \neg y) \wedge z)$.

(a) Find an equivalent formula in CNF without introducing new variables.

Answer:

(b) Draw a circuit representing the original formula (not the one you created in part (a)).

Answer:

(c) Find an equisatisfiable formula in CNF using the Tseitin transformation.

Answer:

5. (40 pts.) Compactness of different encodings of a single function

$PAR_n(x_1, \dots, x_n) = x_1 \oplus \dots \oplus x_n$ is the parity function on n variables, true if and only if the number of true variables is odd (the symbol $\oplus$ is notation for the exclusive OR, a.k.a. XOR, operator). Let's look at how large different representations of this function are: for each of the types of representation below, give an encoding of PAR_n which is as small as possible¹ and compute its size asymptotically (e.g. $\Theta(n^2)$).

(Note: To write conjunctions or disjunctions of an arbitrary number of subformulas, you can use the notation:

$$\bigwedge_{i=1}^k \varphi_i = \varphi_1 \wedge \dots \wedge \varphi_k \quad \bigvee_{i=1}^k \varphi_i = \varphi_1 \vee \dots \vee \varphi_k$$

This is similar to the usual notation $\sum_{i=1}^k f(i)$ for sums. You can also write $\bigwedge_{s \in S} \varphi_s$ if you have a set of subformulas indexed by objects in the set S instead of integers, and you can put constraints on indices in the same way you can when using sums, like $\sum_{\substack{1 \leq i, j \leq k \\ i \neq j}} f(i, j)$.

(a) A DNF (disjunctive normal form, a.k.a. sum-of-products) formula.

Answer:

(b) A propositional formula which can only use the operators $\wedge$, $\vee$, and $\neg$ (this is sometimes called a “Boolean formula”).

(Hint: Try to define PAR_n recursively, in a way that your overall formula won't have exponential size.)

Answer:

(c) An unrestricted propositional formula (i.e. you can use $\rightarrow$ and $\leftrightarrow$).

Answer:

¹You don't have to prove your encoding is (asymptotically) optimal, although it's not too hard to do this for (a), (c), and (d) — feel free to give it a try. On the other hand, the precise lower bound for (b) was a major result!

- (d) A BDD (reduced and ordered; you may pick a variable order).

Answer:

6. (10 pts.) Variable ordering in BDDs

Consider the formula $(x_1 \wedge x_2) \vee (x_3 \wedge x_4)$.

- (a) Construct an ROBDD with the variable order $x_1 > x_2 > x_3 > x_4$.

Answer:

- (b) Construct an ROBDD with the variable order $x_1 > x_3 > x_2 > x_4$.

Answer:

N.B. If you generalize this formula to have n pairs of variables instead of just 2, the first variable order (increasing numbers) will yield a BDD of size $\Theta(n)$, while the second (odd variables first, then even variables) will yield a BDD of size $\Theta(2^n)$. So the choice of variable order really does make a difference in BDDs! If you're interested, you could try to prove these sizes.